Indian Institute of Technology Jammu
Department of Computer Science and Engineering

Natural Language Processing Lab – Assignment 5

Course: Natural Language Processing	Course Code: CS-4-29(TO)
Release Date: _____	Due Date: _____
Maximum Marks: 100	Submission: Google Classroom

Objective

This assignment introduces distributed word representations. You will learn how semantic relationships are captured using dense embeddings and compare different embedding methods.

Goals:

- understand limitations of one-hot representations
- train word embeddings using multiple methods
- analyze semantic properties of embeddings
- visualize embedding spaces

Datasets

Use:

- your dataset from Lab 1 (project corpus)
- AG News for comparison

Part 1: Baseline Representation (10 Marks)

Represent words using:

- One-hot encoding

Discuss:

- dimensionality issues
- sparsity
- lack of semantic similarity

Part 2: Training Word Embeddings (40 Marks)

Train embeddings using:

- CBOW
- Skip-gram
- FastText
- GloVe (library allowed)

Train embeddings with dimensions:

- 80
- 100
- 200

Report:

Method	Dim	Training Time	Vocab Size
CBOW			
Skip-gram			
FastText			
GloVe			

Part 3: Embedding Evaluation (20 Marks)

Evaluate embeddings using:

- cosine similarity between words
- nearest neighbor search
- word analogy tasks (e.g., king - man + woman)

Provide examples showing meaningful semantic relationships.

Part 4: Visualization (15 Marks)

Visualize embeddings using:

- PCA or t-SNE
- TensorFlow Embedding Projector or PyTorch equivalent

Plot:

- clusters of similar words
- semantic groupings

Provide interpretation.

Part 5: Downstream Task (15 Marks)

Use embeddings as features for a task such as:

- text classification

Compare with:

- One-hot / BoW representation

Report:

Representation	Accuracy	F1 Score
BoW		
Embeddings		

Discuss improvements.

Submission

Upload a ZIP file containing:

- source code
- trained embeddings (or link)
- report (PDF)
- plots / visualizations
- instructions to reproduce results

Bonus (Optional, 10 Marks)

- train subword embeddings on Indian language data
- compare OOV handling (FastText vs others)
- evaluate embeddings on similarity benchmarks

Academic Integrity

- code sharing is not allowed
- discussion of ideas is permitted
- external resources must be cited